CONFIDENTIAL | SECURITY OPERATIONS

# Incident Response Report



## MySQL honeypot compromise on Azure VM 'valued-customer'

Aligned to NIST SP 800-61 | Microsoft Defender for Endpoint, Log Analytics & Sentinel

PREPARATION | DETECTION & ANALYSIS | CONTAINMENT & RECOVERY | POST-INCIDENT

| | |
|---|---|
| REPORT DATE | 16 October 2026 |
| INCIDENT WINDOW | 2026-10-08 23:08 UTC to 2026-10-09 ~18:10 UTC |
| AFFECTED ASSET | valued-customer (valued-customers1000), Windows VM, MySQL 8.0 |
| CLASSIFICATION | Simulated incident / honeypot exercise \| Severity: HIGH |
| STATUS | Contained, eradicated and recovered |

# Document control

| Field | Detail |
|---|---|
| Incident reference | IR-2026-001 (exercise placeholder - replace with ticket ID) |
| Prepared by | SOC analyst (name to be inserted) |
| Framework | NIST SP 800-61 incident handling lifecycle (Preparation; Detection & Analysis; Containment, Eradication & Recovery; Post-Incident Activity) |
| Tooling | Microsoft Defender for Endpoint (Advanced Hunting), Azure Monitor Agent + Data Collection Rule, Log Analytics (KQL), Microsoft Sentinel, MySQL Workbench |
| Time convention | All times UTC unless stated. Portal screenshots display local time (EDT, UTC-4). |

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# 1. Executive Summary

| 5+ | 3 | 8 | 0 | 4,854 |
|---|---|---|---|---|
| Distinct external IPs hitting MySQL | Usernames sprayed: root, admin, sa | Attacker login bursts captured | Persistence artifacts found on host | Host events reviewed across 5 tables |

**What happened.** A Windows VM named *valued-customer* was deliberately built as a monitored honeypot: MySQL 8.0 with a weak *root* account, built-in *administrator* and *guest* accounts re-enabled, Windows Firewall disabled and the Azure NSG opened. Exposure began at **2026-10-08T23:08:31Z**. Within hours, automated sources on the public internet found the database service and began credential-guessing. The audit trail shows the same pattern repeating: *root*, *admin* and *sa* are tried within about a second of each other, the *admin* and *sa* attempts fail with "Access denied", and a *root* login then succeeds from the same address.

**Impact.** Unauthorized, authenticated access to the MySQL server as *root* from external IPs is confirmed. Post-login activity in the audit log is limited to client initialisation statements (for example `SELECT @@max_allowed_packet`). Host telemetry (logon, process, file, registry and network events) showed no malware, persistence keys, dropped scripts or suspicious child processes. One block of dump/import-style statements against the *world* schema at 05:14Z could not be attributed and is flagged for verification (see Section 7.4).

**Response.** The investigation package was downloaded, the machine was isolated in Defender for Endpoint, then hardened and returned to service: NSG tightened, Windows Firewall re-enabled, the *administrator* account deleted, *guest* disabled, a strong local account created, public MySQL logon blocked and the network-reachable *root* account secured.

### Overall assessment

**Severity: HIGH.** Initial access was achieved by trivially guessable credentials on an internet-exposed database port. In a production setting this would be a critical, reportable event because the attacker held full database privileges.

**Confirmed:** external brute-force/credential guessing, successful root authentication, repeated activity from the same infrastructure. **Not observed:** persistence, lateral movement, malicious processes. **Unverified:** whether any data was read or altered.

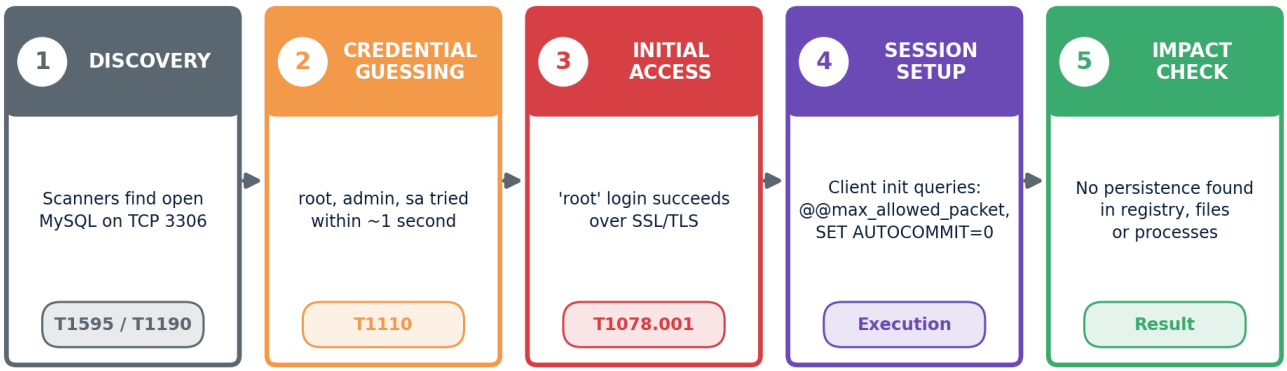



*Figure 1. How the intrusion unfolded, reconstructed from MySQL audit and host telemetry (MITRE ATT&CK references).*

# 2. Scope, Environment and Methodology

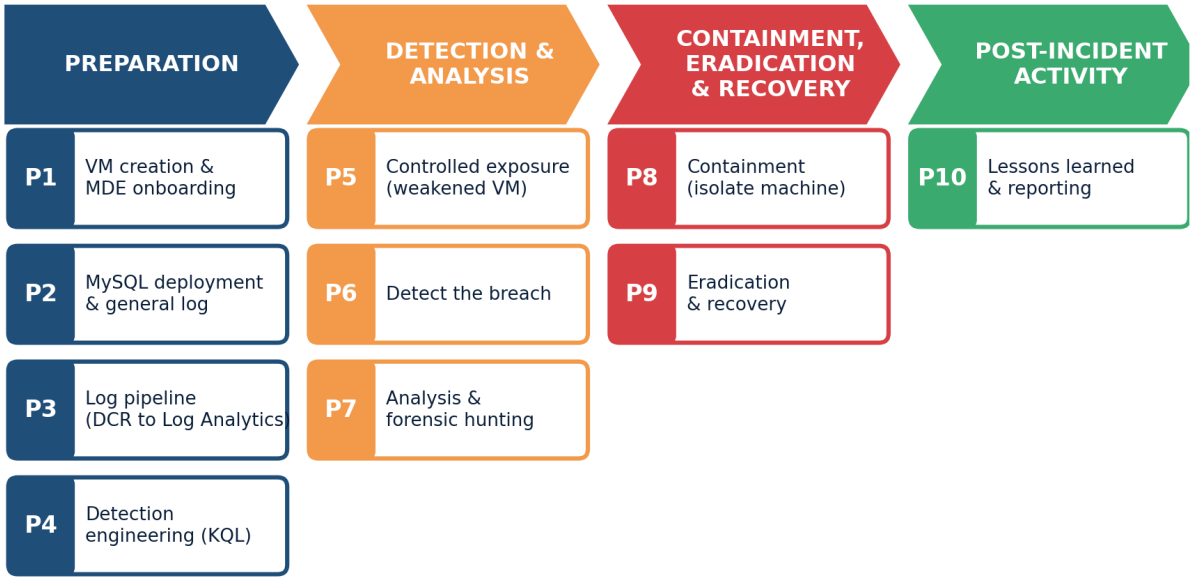



*Figure 2. The ten project phases mapped onto the NIST SP 800-61 incident handling lifecycle.*

The exercise reproduces a realistic cloud-database compromise end to end: build a monitored target, expose it, detect and investigate the intrusion with native Microsoft tooling, contain it, and return the asset to a hardened state. The environment below shows how attacker traffic, host telemetry and database audit logs reach the analyst.

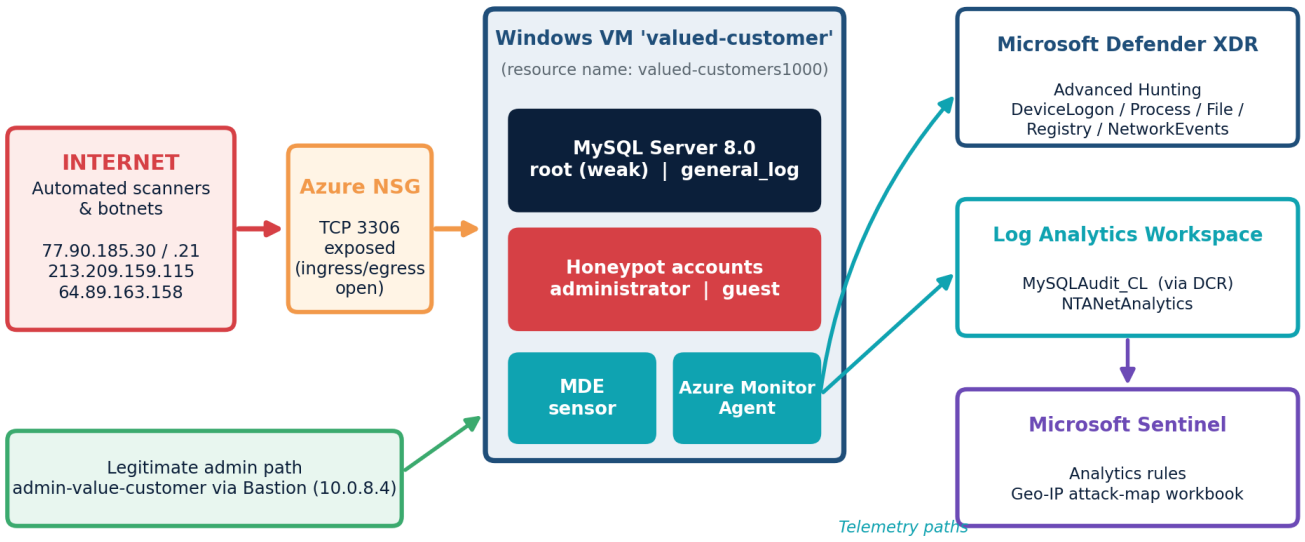



*Figure 3. Environment and telemetry architecture. Red = attacker path, teal = telemetry paths, green = legitimate administration.*

| Item | Value |
|---|---|
| Target host | Windows VM; MDE device name **valued-customer** (MDE truncates to 15 characters); Azure resource / AMA name **valued-customers1000** |
| Service in scope | MySQL Server 8.0 (8.0.45 installer) with general query log enabled (`mysql_general.log`) |
| Telemetry sources | DeviceLogonEvents, DeviceProcessEvents, DeviceFileEvents, DeviceRegistryEvents, DeviceNetworkEvents (MDE); MySQLAudit_CL and NTANetAnalytics (Log Analytics) |
| Exposure start | 2026-10-08T23:08:31.8034371Z (recorded when the VM was weakened; used as the baseline for every hunting query) |
| Method | Detection rules in KQL, post-exposure hunting by table, IP geo-enrichment, Sentinel workbook visualisation, LLM-assisted triage of exported results with analyst review |

Lesson built into the method: **a detection query returns nothing until the activity it looks for actually occurs.** To validate rules before an adversary arrived, authentication telemetry was manufactured on purpose (Section 3.4).

# 3. Preparation (Phases 1-4)

Preparation covers building the asset, wiring its logs to the SIEM and proving the detections work before any exposure. It is the foundation for everything that follows: without these steps the later phases would have no evidence to analyse.

## 3.1 Phase 1 - VM creation and endpoint onboarding

**PHASE 1** **NIST SP 800-61: Preparation**

- Created the Windows VM with the PowerShell hostname **Valued-customer**.
- Onboarded the VM to Microsoft Defender for Endpoint using the Windows 10/11 gateway onboarding package, giving EDR sensor coverage and Advanced Hunting tables.
- **Naming gotcha:** MDE shows the truncated name *valued-customer*, while Azure resources and the Log Analytics `_ResourceId` carry *valued-customers1000*. Queries use `startswith` or the exact name for the right table.

## 3.2 Phase 2 - MySQL deployment

**PHASE 2** **NIST SP 800-61: Preparation**

- Installed the Visual C++ x64 redistributable and MySQL Installer Community 8.0.45.
- Configured the *root* account and populated the sample database (the *world* schema seen later in the audit log).
- Applied a custom `my.ini` that enables the general query log; restarted the service from `services.msc` and confirmed `general_log` was active.
- Cloned the database using MySQL Workbench Data Export/Import (relevant to the finding in Section 7.4).

## 3.3 Phase 3 - Wiring logs to Log Analytics

**PHASE 3** **NIST SP 800-61: Preparation**

A Data Collection Rule (DCR) tells Azure Monitor where to read the MySQL log and how to parse it:

| DCR setting | Value |
|---|---|
| File pattern | C:\ProgramData\MySQL\MySQL Server 8.0\Data\mysql_general.log |
| Destination table | MySQLAudit_CL (custom log table) |
| Record delimiter | TimeStamp |
| Timestamp format | ISO 8601 |

## 3.4 Phase 4 - Detection engineering

**PHASE 4** **NIST SP 800-61: Preparation**

Two scheduled detections were written. The first watches built-in accounts on the host; the second reconstructs MySQL logon success and failure from raw general-log lines by pairing connection IDs.

| Rule name | Data source | Logic |
|---|---|---|
| **Valued_VM_Login** | DeviceLogonEvents | Successful logon (`LogonSuccess`) by *administrator* or *guest* on the VM. |

| Rule name | Data source | Logic |
|---|---|---|
| **Valued_MYSQLSERVER_Login** | MySQLAudit_CL | Extracts ConnectionId; any connection that also produced "Access denied" becomes *LogonFailure*, other connects become *LogonSuccess*; parses Username and IpAddress. |

**Rule: Valued_VM_Login**

```
let MyDevice = "valued-customers1000";   // MDE truncates the device name
DeviceLogonEvents
| where DeviceName == MyDevice
| where AccountName in~ ("administrator", "guest")
| where ActionType == "LogonSuccess"
| project TimeGenerated, RemoteIP, AccountName, DeviceName, ActionType, LogonType
```

**Rule: Valued_MYSQLSERVER_Login**

```
let MyDevice = "valued-customers1000";  let MyTimeframe = ago(7d);
let FailedConnections = MySQLAudit_CL
  | where TimeGenerated > MyTimeframe
  | extend RawData = replace_string(RawData, "\t", " ")
  | extend DeviceName = tostring(split(_ResourceId, "/")[-1])
  | where DeviceName startswith MyDevice and RawData has "Access denied"
  | extend ConnectionId = extract(@"\s+(\d+)\s+Access denied", 1, RawData)
  | where isnotempty(ConnectionId) | distinct ConnectionId;
MySQLAudit_CL
| where TimeGenerated > MyTimeframe
| extend RawData = replace_string(RawData, "\t", " ")
| extend DeviceName = tostring(split(_ResourceId, "/")[-1])
| where DeviceName startswith MyDevice
| where RawData has "Connect" or RawData has "Access denied"
| extend ConnectionId = extract(@"\s+(\d+)\s+(?:Connect|Access denied)", 1, RawData)
| where isnotempty(ConnectionId)
| extend ActionType = case(RawData has "Access denied", "LogonFailure",
                           ConnectionId in (FailedConnections), "Ignore", "LogonSuccess")
| where ActionType != "Ignore"
| extend Username  = replace_string(tostring(split(tostring(split(RawData,"@")[0])," ")[-1]),"'","")
| extend IpAddress = replace_string(tostring(split(tostring(split(RawData,"@")[1])," ")[0]),"'","")
| project TimeGenerated, DeviceName, Username, IpAddress, ActionType, RawData
| order by TimeGenerated desc
```

## 3.5 Validating the detection - manufacturing telemetry

With no real attacker yet, the *Valued_VM_Login* query returned zero rows. This is the **zero-result rule**: a rule targeting high-risk accounts stays silent until someone actually authenticates against them. To prove the rule worked, logon attempts were generated on the VM with a credential prompt (any password, or blank, then OK) and the query was re-run after 1-2 minutes for MDE ingestion.

**PowerShell used to simulate authentication against the honeypot accounts**

```
Start-Process cmd.exe -Credential (Get-Credential "administrator")
Start-Process cmd.exe -Credential (Get-Credential "guest")
```

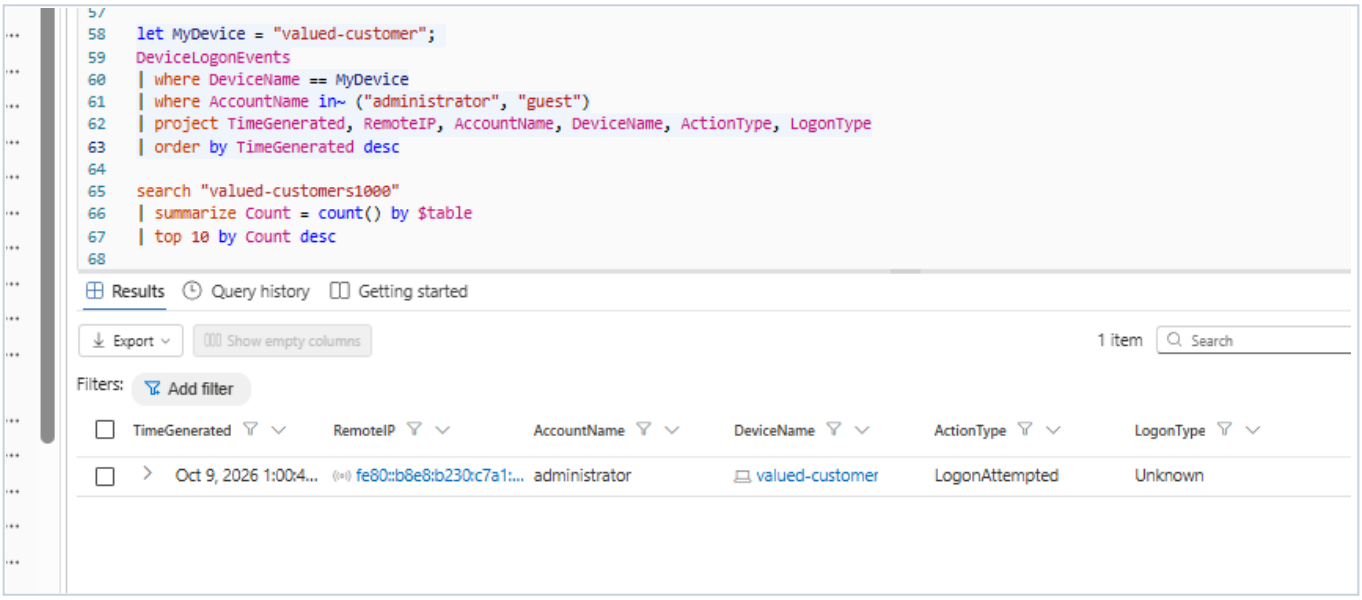



*Figure 4. First hit: the administrator simulation appears in DeviceLogonEvents as a LogonAttempted event.*

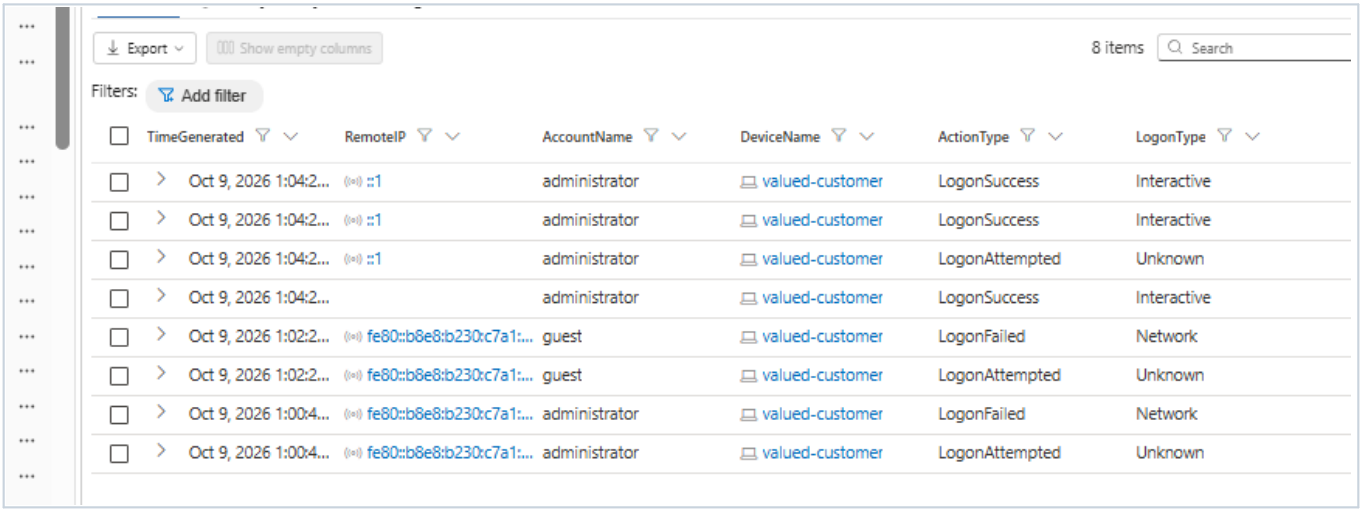



*Figure 5. After both simulations: eight events for administrator and guest (LogonAttempted, LogonFailed, LogonSuccess), confirming the detection logic works.*

### Key learning

Detection rules can be validated safely by simulating the behaviour they target. The simulated events land as *LogonAttempted*, *LogonFailed* or *LogonSuccess* depending on how Windows evaluates the credential, so rule logic should account for all three action types.

# 4. Detection and Analysis (Phases 5-7)

## 4.1 Phase 5 - Controlled exposure

PHASE 5 NIST SP 800-61: Detection & Analysis

To invite attack, the VM was intentionally weakened. The exact moment was recorded because it becomes the baseline for every later query.

| Control weakened | How |
|---|---|
| Built-in accounts | *administrator* and *guest* added/enabled in `compmgmt.msc`; guest password weakened |
| Local security policy | Relaxed via `secpol.msc` |
| Host firewall | Windows Firewall disabled |
| Network | Azure NSG opened so inbound and egress traffic is allowed (MySQL reachable on 3306) |
| Database | Network-reachable MySQL *root* account with a weak password |
| Evidence | Investigation package kept for later comparison |

**Baseline timestamp**

**`2026-10-08T23:08:31.8034371Z`** - start of exposure; all hunting queries filter on TimeGenerated >= this value.

## 4.2 Phase 6 - Detecting the breach: MySQL logons

PHASE 6 NIST SP 800-61: Detection & Analysis

Re-running the MySQL logon query against the baseline timestamp immediately surfaced the intrusion. Results are grouped in repeating bursts, newest first.

| TimeGenerated | DeviceName | Username | IpAddress | ActionType | RawData |
|---|---|---|---|---|---|
| Oct 9, 2026 2:10:2... | valued-customers1000 | | | LogonSuccess | 2026-10-09T18:10:26.88... |
| Oct 9, 2026 1:40:1... | valued-customers1000 | root | 77.90.185.30 | LogonSuccess | 2026-10-09T17:40:10.21... |
| Oct 9, 2026 1:40:0... | valued-customers1000 | sa | 77.90.185.30 | LogonFailure | 2026-10-09T17:40:09.64... |
| Oct 9, 2026 1:40:0... | valued-customers1000 | admin | 77.90.185.30 | LogonFailure | 2026-10-09T17:40:08.97... |
| Oct 9, 2026 1:40:0... | valued-customers1000 | root | 77.90.185.30 | LogonFailure | 2026-10-09T17:40:08.40... |
| Oct 9, 2026 1:05:1... | valued-customers1000 | sa | 77.90.185.21 | LogonFailure | 2026-10-09T17:05:18.21... |
| Oct 9, 2026 1:05:1... | valued-customers1000 | root | 77.90.185.21 | LogonSuccess | 2026-10-09T17:05:18.78... |
| Oct 9, 2026 1:05:1... | valued-customers1000 | admin | 77.90.185.21 | LogonFailure | 2026-10-09T17:05:17.53... |
| Oct 9, 2026 1:05:1... | valued-customers1000 | root | 77.90.185.21 | LogonFailure | 2026-10-09T17:05:16.96... |
| Oct 9, 2026 12:24:... | valued-customers1000 | root | 77.90.185.30 | LogonSuccess | 2026-10-09T16:24:11.67... |
| Oct 9, 2026 12:24:... | valued-customers1000 | sa | 77.90.185.30 | LogonFailure | 2026-10-09T16:24:10.99... |
| Oct 9, 2026 12:24:... | valued-customers1000 | admin | 77.90.185.30 | LogonFailure | 2026-10-09T16:24:10.30... |
| Oct 9, 2026 12:24:... | valued-customers1000 | root | 77.90.185.30 | LogonFailure | 2026-10-09T16:24:09.73... |
| Oct 9, 2026 3:40:2... | valued-customers1000 | root | 64.89.163.158 | LogonSuccess | 2026-10-09T07:40:23.61... |
| Oct 9, 2026 3:40:2... | valued-customers1000 | root | 64.89.163.158 | LogonFailure | 2026-10-09T07:40:20.33... |
| Oct 9, 2026 3:40:1... | valued-customers1000 | | | LogonSuccess | 2026-10-09T07:40:16.63... |
| Oct 9, 2026 3:29:2... | valued-customers1000 | root | 77.90.185.30 | LogonSuccess | 2026-10-09T07:29:26.21... |
| Oct 9, 2026 3:29:2... | valued-customers1000 | sa | 77.90.185.30 | LogonFailure | 2026-10-09T07:29:25.64... |
| Oct 9, 2026 3:29:2... | valued-customers1000 | admin | 77.90.185.30 | LogonFailure | 2026-10-09T07:29:24.96... |
| Oct 9, 2026 3:29:2... | valued-customers1000 | root | 77.90.185.30 | LogonFailure | 2026-10-09T07:29:24.41... |
| Oct 9, 2026 2:11:2... | valued-customers1000 | root | 77.90.185.30 | LogonSuccess | 2026-10-09T06:11:25.67... |
| Oct 9, 2026 2:11:2... | valued-customers1000 | sa | 77.90.185.30 | LogonFailure | 2026-10-09T06:11:25.11... |
| Oct 9, 2026 2:11:2... | valued-customers1000 | admin | 77.90.185.30 | LogonFailure | 2026-10-09T06:11:24.43... |
| Oct 9, 2026 2:11:2... | valued-customers1000 | root | 77.90.185.30 | LogonFailure | 2026-10-09T06:11:23.85... |
| Oct 9, 2026 1:59:3... | valued-customers1000 | root | 213.209.159.115 | LogonSuccess | 2026-10-09T05:59:33.03... |
| Oct 9, 2026 1:59:3... | valued-customers1000 | admin | 213.209.159.115 | LogonFailure | 2026-10-09T05:59:32.00... |

*Figure 6. Query 1 results - MySQL LogonFailure and LogonSuccess events. Failures for admin/sa/root precede a root success from the same IP.*

- **Signature of automation.** Within milliseconds, one source IP tries *root*, *admin* and *sa*. Failures are logged as LogonFailure ("Access denied"); a *root* LogonSuccess follows from the same IP.
- **Repeating infrastructure.** 77.90.185.30 and 77.90.185.21 return again and again across the window; 213.209.159.115 and 64.89.163.158 appear as additional sources, and 45.128.199.171 appears in the exported audit dataset.
- **Consistent cadence.** Bursts recur at regular intervals - another hallmark of tooling rather than a human operator.
- **Target choice.** *root* is the preferred target because it holds ultimate privilege on MySQL.

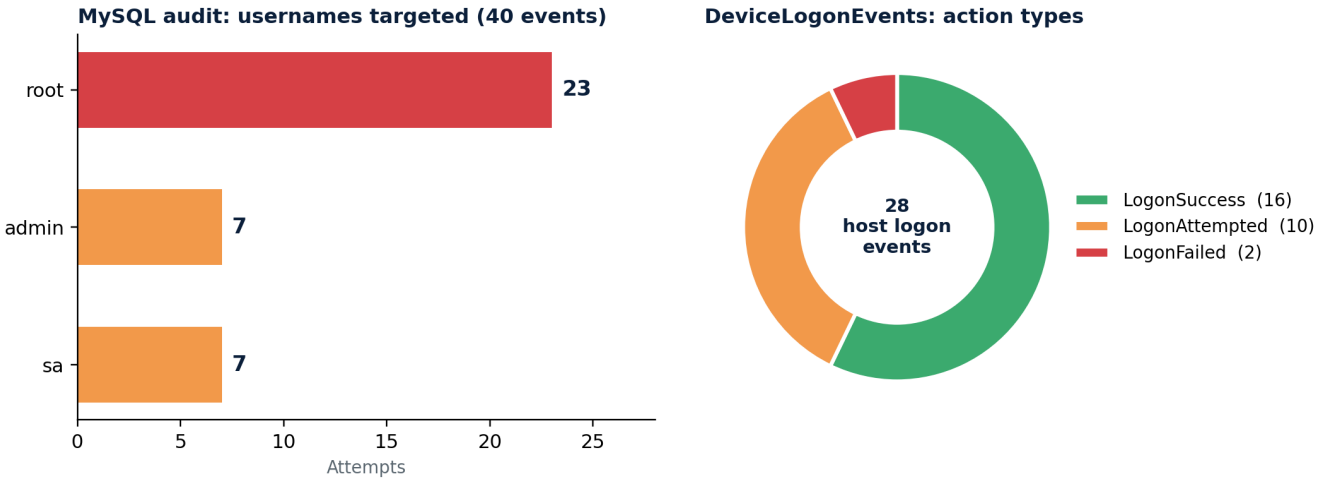



*Figure 7. Left: username targeting in the 40-event MySQL audit export. Right: breakdown of the 28 host logon events.*

## 4.3 Phase 7 - Analysis: what did the attacker do after login?

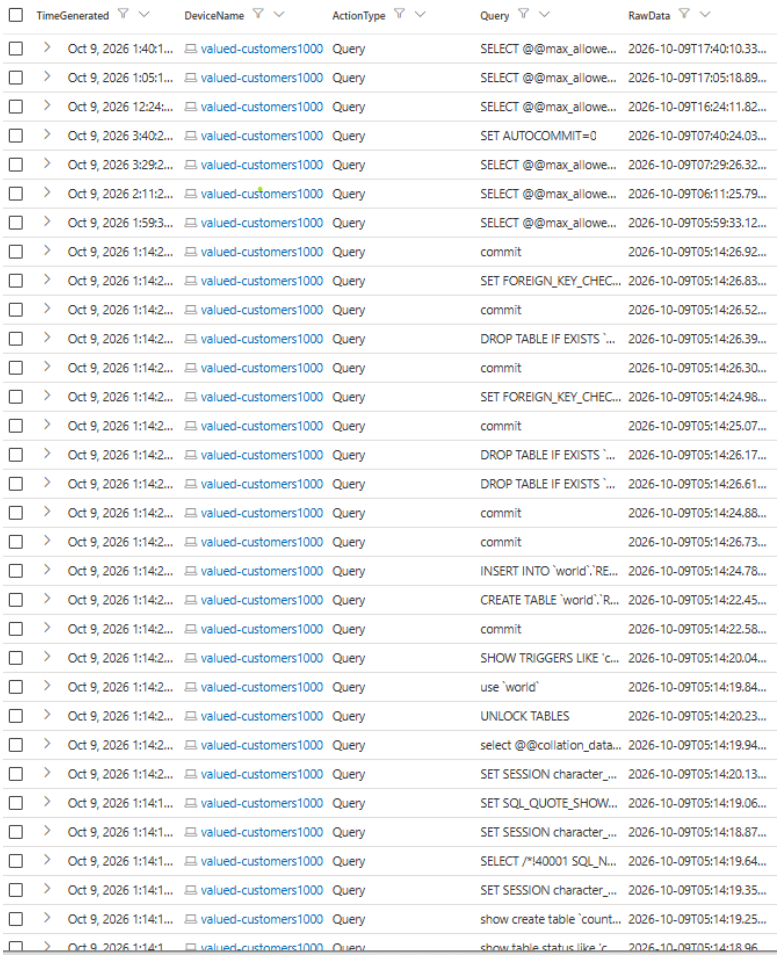
**PHASE 7** NIST SP 800-61: Detection & Analysis

Phase 7 saved the query results and analysed them, first the database layer, then the host, then the network. Query results were exported and triaged with LLM assistance; the interpretation below reflects that review plus analyst corrections.

### 4.3.1 MySQL query tracking

**Query 2 (condensed)**

```
MySQLAudit_CL
| where TimeGenerated > ServerVulnerableDateTime
| extend RawData = replace_string(RawData, "\t", " ")
| extend DeviceName = tostring(split(_ResourceId, "/")[-1])
| where DeviceName startswith MyDevice and RawData has "Query"
| extend Query = trim(" ", tostring(split(RawData, "Query")[-1]))
| project TimeGenerated, DeviceName, ActionType = "Query", Query, RawData
| order by TimeGenerated desc
```



| TimeGenerated | DeviceName | ActionType | Query | RawData |
|---|---|---|---|---|
| Oct 9, 2026 1:40:1... | valued-customers1000 | Query | SELECT @@max_allowe... | 2026-10-09T17:40:10.33... |
| Oct 9, 2026 1:05:1... | valued-customers1000 | Query | SELECT @@max_allowe... | 2026-10-09T17:05:18.89... |
| Oct 9, 2026 12:24:... | valued-customers1000 | Query | SELECT @@max_allowe... | 2026-10-09T16:24:11.82... |
| Oct 9, 2026 3:40:2... | valued-customers1000 | Query | SET AUTOCOMMIT=0 | 2026-10-09T07:40:24.03... |
| Oct 9, 2026 3:29:2... | valued-customers1000 | Query | SELECT @@max_allowe... | 2026-10-09T07:29:26.32... |
| Oct 9, 2026 2:11:2... | valued-customers1000 | Query | SELECT @@max_allowe... | 2026-10-09T06:11:25.79... |
| Oct 9, 2026 1:59:3... | valued-customers1000 | Query | SELECT @@max_allowe... | 2026-10-09T05:59:33.12... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:26.92... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | SET FOREIGN_KEY_CHEC... | 2026-10-09T05:14:26.83... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:26.52... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | DROP TABLE IF EXISTS `... | 2026-10-09T05:14:26.39... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:26.30... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | SET FOREIGN_KEY_CHEC... | 2026-10-09T05:14:24.98... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:25.07... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | DROP TABLE IF EXISTS `... | 2026-10-09T05:14:26.17... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | DROP TABLE IF EXISTS `... | 2026-10-09T05:14:26.61... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:24.88... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:26.73... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | INSERT INTO `world`.`RE... | 2026-10-09T05:14:24.78... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | CREATE TABLE `world`.`R... | 2026-10-09T05:14:22.45... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | commit | 2026-10-09T05:14:22.58... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | SHOW TRIGGERS LIKE 'c... | 2026-10-09T05:14:20.04... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | use `world` | 2026-10-09T05:14:19.84... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | UNLOCK TABLES | 2026-10-09T05:14:20.23... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | select @@collation_data... | 2026-10-09T05:14:19.94... |
| Oct 9, 2026 1:14:2... | valued-customers1000 | Query | SET SESSION character_... | 2026-10-09T05:14:20.13... |
| Oct 9, 2026 1:14:1... | valued-customers1000 | Query | SET SQL_QUOTE_SHOW... | 2026-10-09T05:14:19.06... |
| Oct 9, 2026 1:14:1... | valued-customers1000 | Query | SET SESSION character_... | 2026-10-09T05:14:18.87... |
| Oct 9, 2026 1:14:1... | valued-customers1000 | Query | SELECT /*!40001 SQL_N... | 2026-10-09T05:14:19.64... |
| Oct 9, 2026 1:14:1... | valued-customers1000 | Query | SET SESSION character_... | 2026-10-09T05:14:19.35... |
| Oct 9, 2026 1:14:1... | valued-customers1000 | Query | show create table `count... | 2026-10-09T05:14:19.25... |
| Oct 9, 2026 1:14:1... | valued-customers1000 | Query | show table status like 'c... | 2026-10-09T05:14:18.96... |

*Figure 8. Query 2 results - statements executed after exposure.*

**Session initialisation.** After each attacker login the log shows `SELECT @@max_allowed_packet` (and at 07:40Z `SET AUTOCOMMIT=0`), typical first statements sent by database clients and scripts when a session opens.

**Schema block at 05:14Z.** A tight cluster shows `use world`, `SHOW TRIGGERS`, `SET FOREIGN_KEY_CHECKS=0`, `DROP TABLE IF EXISTS`, `CREATE TABLE`, `INSERT INTO`, `UNLOCK TABLES` and `COMMIT`.

**Assessment.** The original triage read this as an attacker toolkit. That sequence is also exactly what MySQL Workbench emits when a dump file is imported (the Phase 2 clone). It precedes the first captured attacker login (05:59Z) and carries no external IP in the captured rows, so it is recorded here as **unattributed - verify** rather than as confirmed malicious activity.

### 4.3.2 Honeypot logons on the host

**Query 3**

```
DeviceLogonEvents
| where TimeGenerated > ServerVulnerableDateTime and DeviceName == MyDevice
| where AccountName in~ ("administrator", "guest")
| project TimeGenerated, RemoteIP, AccountName, DeviceName, ActionType, LogonType
```

Filters: Add filter

| TimeGenerated | RemoteIP | AccountName | DeviceName | ActionType | LogonType |
|---|---|---|---|---|---|
| Oct 9, 2026 1:00:4... | fe80::b8e8:b230:c7a1:... | administrator | valued-customer | LogonAttempted | Unknown |
| Oct 9, 2026 1:00:4... | fe80::b8e8:b230:c7a1:... | administrator | valued-customer | LogonFailed | Network |
| Oct 9, 2026 1:02:2... | fe80::b8e8:b230:c7a1:... | guest | valued-customer | LogonAttempted | Unknown |
| Oct 9, 2026 1:02:2... | fe80::b8e8:b230:c7a1:... | guest | valued-customer | LogonFailed | Network |
| Oct 9, 2026 1:04:2... | | administrator | valued-customer | LogonSuccess | Interactive |
| Oct 9, 2026 1:04:2... | ::1 | administrator | valued-customer | LogonSuccess | Interactive |
| Oct 9, 2026 1:04:2... | ::1 | administrator | valued-customer | LogonAttempted | Unknown |
| Oct 9, 2026 1:04:2... | ::1 | administrator | valued-customer | LogonSuccess | Interactive |
| Oct 9, 2026 1:10:3... | | guest | valued-customer | LogonSuccess | Interactive |
| Oct 9, 2026 1:10:3... | ::1 | guest | valued-customer | LogonAttempted | Unknown |
| Oct 9, 2026 1:10:3... | ::1 | guest | valued-customer | LogonSuccess | Interactive |

*Figure 9. Query 3 results - honeypot account activity after exposure.*

- **Network-level probes:** first rows show LogonAttempted then LogonFailed (LogonType Network) from a link-local IPv6 address for *administrator* and *guest* - rejected.
- **Local validation:** later rows from `::1` or a blank address are Interactive logons, matching the PowerShell credential simulations. This separates operator testing from hostile activity.

### 4.3.3 Post-compromise host hunting

Every table was queried from the baseline timestamp forward. The pattern is the same for each: project the useful columns, order newest first, and look for anything that is not routine OS, Azure agent or EDR behaviour.

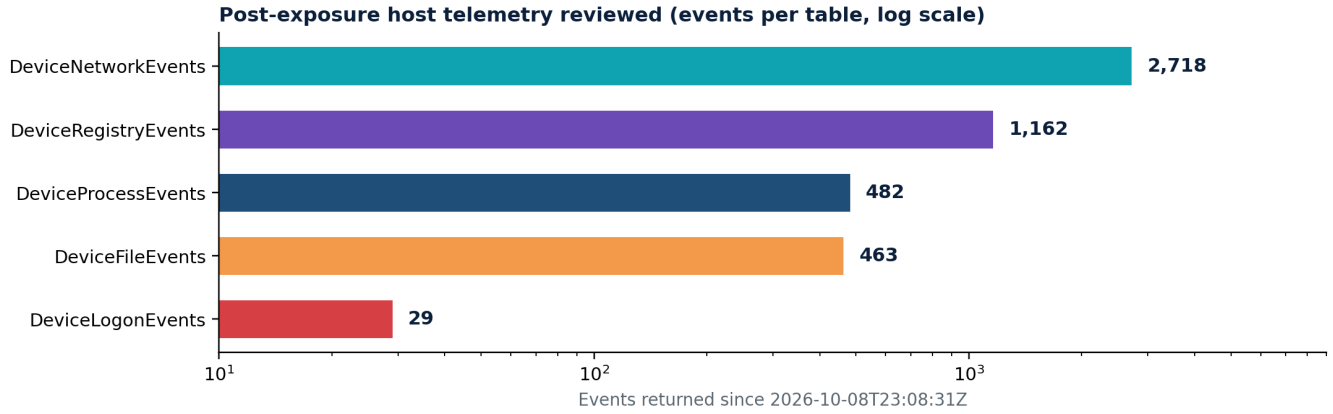
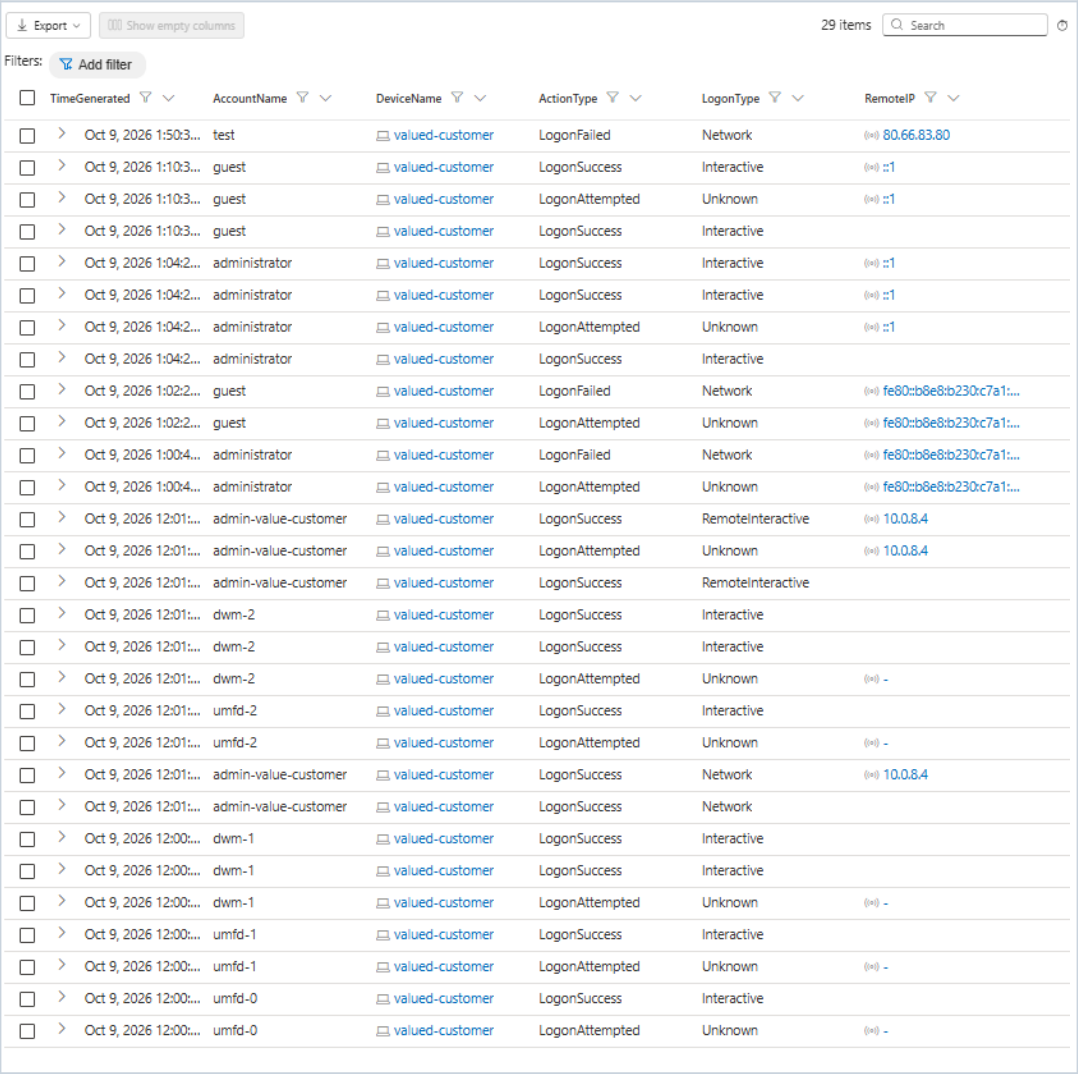



*Figure 10. Volume of host telemetry reviewed per table.*

**Logon events (29 rows).** Beyond the simulations, the table holds a failed Network logon for account *test* from 80.66.83.80 (spraying of an arbitrary username), legitimate RemoteInteractive sessions by *admin-value-customer* from 10.0.8.4 (Bastion/private path), and standard Windows session accounts (*dwm-**, *umfd-**).



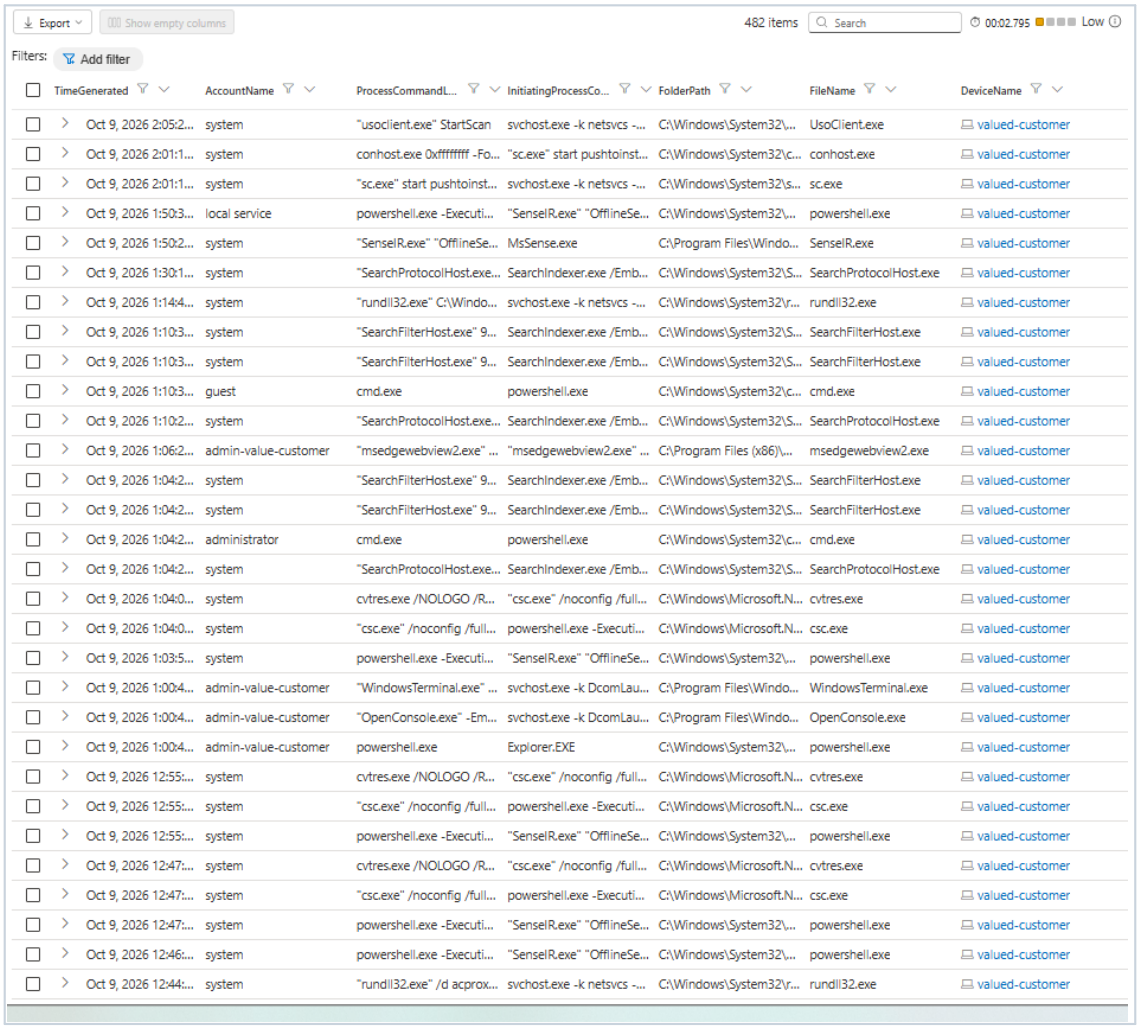
Export | Show empty columns | 29 items | Search

Filters: Add filter

| TimeGenerated | AccountName | DeviceName | ActionType | LogonType | RemoteIP |
|---|---|---|---|---|---|
| Oct 9, 2026 1:50:3... | test | valued-customer | LogonFailed | Network | 80.66.83.80 |
| Oct 9, 2026 1:10:3... | guest | valued-customer | LogonSuccess | Interactive | ::1 |
| Oct 9, 2026 1:10:3... | guest | valued-customer | LogonAttempted | Unknown | ::1 |
| Oct 9, 2026 1:10:3... | guest | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 1:04:2... | administrator | valued-customer | LogonSuccess | Interactive | ::1 |
| Oct 9, 2026 1:04:2... | administrator | valued-customer | LogonSuccess | Interactive | ::1 |
| Oct 9, 2026 1:04:2... | administrator | valued-customer | LogonAttempted | Unknown | ::1 |
| Oct 9, 2026 1:04:2... | administrator | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 1:02:2... | guest | valued-customer | LogonFailed | Network | fe80::b8e8:b230:c7a1:... |
| Oct 9, 2026 1:02:2... | guest | valued-customer | LogonAttempted | Unknown | fe80::b8e8:b230:c7a1:... |
| Oct 9, 2026 1:00:4... | administrator | valued-customer | LogonFailed | Network | fe80::b8e8:b230:c7a1:... |
| Oct 9, 2026 1:00:4... | administrator | valued-customer | LogonAttempted | Unknown | fe80::b8e8:b230:c7a1:... |
| Oct 9, 2026 12:01:... | admin-value-customer | valued-customer | LogonSuccess | RemoteInteractive | 10.0.8.4 |
| Oct 9, 2026 12:01:... | admin-value-customer | valued-customer | LogonAttempted | Unknown | 10.0.8.4 |
| Oct 9, 2026 12:01:... | admin-value-customer | valued-customer | LogonSuccess | RemoteInteractive | |
| Oct 9, 2026 12:01:... | dwm-2 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:01:... | dwm-2 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:01:... | dwm-2 | valued-customer | LogonAttempted | Unknown | - |
| Oct 9, 2026 12:01:... | umfd-2 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:01:... | umfd-2 | valued-customer | LogonAttempted | Unknown | - |
| Oct 9, 2026 12:01:... | admin-value-customer | valued-customer | LogonSuccess | Network | 10.0.8.4 |
| Oct 9, 2026 12:01:... | admin-value-customer | valued-customer | LogonSuccess | Network | |
| Oct 9, 2026 12:00:... | dwm-1 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:00:... | dwm-1 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:00:... | dwm-1 | valued-customer | LogonAttempted | Unknown | - |
| Oct 9, 2026 12:00:... | umfd-1 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:00:... | umfd-1 | valued-customer | LogonAttempted | Unknown | - |
| Oct 9, 2026 12:00:... | umfd-0 | valued-customer | LogonSuccess | Interactive | |
| Oct 9, 2026 12:00:... | umfd-0 | valued-customer | LogonAttempted | Unknown | - |

*Figure 11. DeviceLogonEvents since exposure (29 items).*

**Process events (482 rows).** The cmd.exe launches under *guest* and *administrator* (parent powershell.exe) are the operator simulations. The bulk of activity is system context: Defender SenseIR collection scripts, UsoClient, sc.exe and search indexer processes. No suspicious child process, encoded command or download cradle was seen.



Export | Show empty columns | 482 items | Search | 00:02.795 | Low

Filters: Add filter

| TimeGenerated | AccountName | ProcessCommandL... | InitiatingProcessCo... | FolderPath | FileName | DeviceName |
|---|---|---|---|---|---|---|
| Oct 9, 2026 2:05:2... | system | "usoclient.exe" StartScan | svchost.exe -k netsvcs -... | C:\Windows\System32\... | UsoClient.exe | valued-customer |
| Oct 9, 2026 2:01:1... | system | conhost.exe 0xffffffff -Fo... | "sc.exe" start pushtoinst... | C:\Windows\System32\c... | conhost.exe | valued-customer |
| Oct 9, 2026 2:01:1... | system | "sc.exe" start pushtoinst... | svchost.exe -k netsvcs -... | C:\Windows\System32\s... | sc.exe | valued-customer |
| Oct 9, 2026 1:50:3... | local service | powershell.exe -Executi... | "SenseIR.exe" "OfflineSe... | C:\Windows\System32\... | powershell.exe | valued-customer |
| Oct 9, 2026 1:50:2... | system | "SenseIR.exe" "OfflineSe... | MsSense.exe | C:\Program Files\Windo... | SenseIR.exe | valued-customer |
| Oct 9, 2026 1:30:1... | system | "SearchProtocolHost.exe... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchProtocolHost.exe | valued-customer |
| Oct 9, 2026 1:14:4... | system | "rundll32.exe" C:\Windo... | svchost.exe -k netsvcs -... | C:\Windows\System32\r... | rundll32.exe | valued-customer |
| Oct 9, 2026 1:10:3... | system | "SearchFilterHost.exe" 9... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchFilterHost.exe | valued-customer |
| Oct 9, 2026 1:10:3... | system | "SearchFilterHost.exe" 9... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchFilterHost.exe | valued-customer |
| Oct 9, 2026 1:10:3... | guest | cmd.exe | powershell.exe | C:\Windows\System32\c... | cmd.exe | valued-customer |
| Oct 9, 2026 1:10:2... | system | "SearchProtocolHost.exe... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchProtocolHost.exe | valued-customer |
| Oct 9, 2026 1:06:2... | admin-value-customer | "msedgewebview2.exe" ... | "msedgewebview2.exe" ... | C:\Program Files (x86)\... | msedgewebview2.exe | valued-customer |
| Oct 9, 2026 1:04:2... | system | "SearchFilterHost.exe" 9... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchFilterHost.exe | valued-customer |
| Oct 9, 2026 1:04:2... | system | "SearchFilterHost.exe" 9... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchFilterHost.exe | valued-customer |
| Oct 9, 2026 1:04:2... | administrator | cmd.exe | powershell.exe | C:\Windows\System32\c... | cmd.exe | valued-customer |
| Oct 9, 2026 1:04:2... | system | "SearchProtocolHost.exe... | SearchIndexer.exe /Emb... | C:\Windows\System32\S... | SearchProtocolHost.exe | valued-customer |
| Oct 9, 2026 1:04:0... | system | cvtres.exe /NOLOGO /R... | "csc.exe" /noconfig /full... | C:\Windows\Microsoft.N... | cvtres.exe | valued-customer |
| Oct 9, 2026 1:04:0... | system | "csc.exe" /noconfig /full... | powershell.exe -Executi... | C:\Windows\Microsoft.N... | csc.exe | valued-customer |
| Oct 9, 2026 1:03:5... | system | powershell.exe -Executi... | "SenseIR.exe" "OfflineSe... | C:\Windows\System32\... | powershell.exe | valued-customer |
| Oct 9, 2026 1:00:4... | admin-value-customer | "WindowsTerminal.exe" ... | svchost.exe -k DcomLau... | C:\Program Files\Windo... | WindowsTerminal.exe | valued-customer |
| Oct 9, 2026 1:00:4... | admin-value-customer | "OpenConsole.exe" -Em... | svchost.exe -k DcomLau... | C:\Program Files\Windo... | OpenConsole.exe | valued-customer |
| Oct 9, 2026 1:00:4... | admin-value-customer | powershell.exe | Explorer.EXE | C:\Windows\System32\... | powershell.exe | valued-customer |
| Oct 9, 2026 12:55:... | system | cvtres.exe /NOLOGO /R... | "csc.exe" /noconfig /full... | C:\Windows\Microsoft.N... | cvtres.exe | valued-customer |
| Oct 9, 2026 12:55:... | system | "csc.exe" /noconfig /full... | powershell.exe -Executi... | C:\Windows\Microsoft.N... | csc.exe | valued-customer |
| Oct 9, 2026 12:55:... | system | powershell.exe -Executi... | "SenseIR.exe" "OfflineSe... | C:\Windows\System32\... | powershell.exe | valued-customer |
| Oct 9, 2026 12:47:... | system | cvtres.exe /NOLOGO /R... | "csc.exe" /noconfig /full... | C:\Windows\Microsoft.N... | cvtres.exe | valued-customer |
| Oct 9, 2026 12:47:... | system | "csc.exe" /noconfig /full... | powershell.exe -Executi... | C:\Windows\Microsoft.N... | csc.exe | valued-customer |
| Oct 9, 2026 12:47:... | system | powershell.exe -Executi... | "SenseIR.exe" "OfflineSe... | C:\Windows\System32\... | powershell.exe | valued-customer |
| Oct 9, 2026 12:46:... | system | powershell.exe -Executi... | "SenseIR.exe" "OfflineSe... | C:\Windows\System32\... | powershell.exe | valued-customer |
| Oct 9, 2026 12:44:... | system | "rundll32.exe" /d acprox... | svchost.exe -k netsvcs -... | C:\Windows\System32\r... | rundll32.exe | valued-customer |

*Figure 12. DeviceProcessEvents since exposure (482 items).*

**File events (463 rows).** A cluster of FileRenamed events under `C:\Users\Guest\AppData` at the time of the guest simulation is first-logon profile initialisation (Start Menu and WinX shortcuts). Other entries are PowerShell script-policy test files and Azure agent log packaging. No scripts or binaries were dropped in startup or temp locations.

Filters: Add filter

| TimeGenerated | InitiatingProcessAc... | FolderPath | FileName | ActionType | DeviceName |
|---|---|---|---|---|---|
| Oct 9, 2026 2:02:1... | system | C:\WindowsAzure\Colle... | VMAgentLogs.zip | FileModified | valued-customer |
| Oct 9, 2026 2:01:2... | system | C:\Windows\apppatch\... | 01DA54B25CE0D000.ms... | FileCreated | valued-customer |
| Oct 9, 2026 2:01:2... | system | C:\Windows\apppatch\... | 01DA54B25CE0D000.sys... | FileCreated | valued-customer |
| Oct 9, 2026 1:50:3... | local service | C:\Windows\ServiceProfi... | __PSScriptPolicyTest_a5x... | FileCreated | valued-customer |
| Oct 9, 2026 1:50:3... | local service | C:\Windows\ServiceProfi... | __PSScriptPolicyTest_ok0... | FileCreated | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 1 - Desktop.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 1 - Run.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 2 - Search.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 3 - Windows Explorer.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 4 - Control Panel.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 5 - Task Manager.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 01a - Windows PowerSh... | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 03 - Computer Manage... | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 02a - Windows PowerSh... | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 04 - Disk Management.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 04-1 - NetworkStatus.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 05 - Device Manager.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 06 - SystemAbout.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 08 - PowerAndSleep.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 07 - Event Viewer.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 09 - Mobility Center.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | 10 - AppsAndFeatures.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | Shows Desktop.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | Window Switcher.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | LiveCaptions.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | Magnify.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | Narrator.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | On-Screen Keyboard.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | VoiceAccess.lnk | FileRenamed | valued-customer |
| Oct 9, 2026 1:10:3... | system | C:\Users\Guest\AppData... | Administrative Tools.lnk | FileRenamed | valued-customer |

*Figure 13. DeviceFileEvents since exposure.*

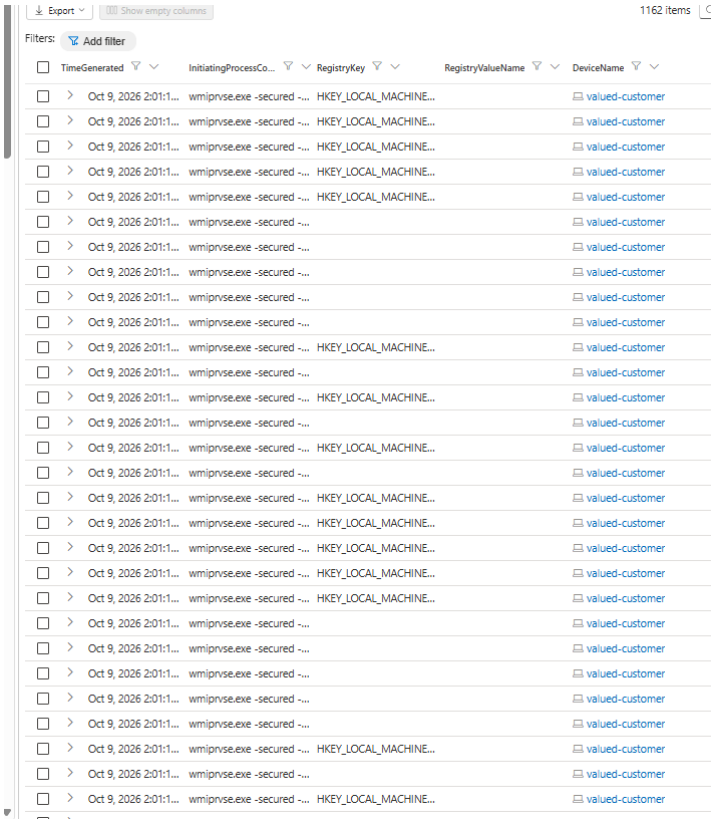



*Figure 14. DeviceRegistryEvents since exposure (1,162 items).*

**Registry events (1,162 rows).** Activity is dominated by `wmiprvse.exe -secured -Embedding` reading HKLM and Boot Configuration Data - routine for an Azure VM, driven by the guest agent and monitoring extensions.

Also present: `wevtutil.exe` registering ETW manifests for Azure extensions and Defender, and Background Activity Moderator (BAM) entries tied to user SIDs.

**Result:** no unauthorized Run, RunOnce, service or Image File Execution Options changes were seen in the reviewed events.

**Network events (2,718 rows).** Traffic comprises Azure infrastructure (168.63.129.16 DNS/metadata), internal gateway 10.3.0.33, Microsoft service endpoints on 443 (OneDrive, Defender, telemetry) and mysqld.exe sessions from public addresses. Individual outbound destinations are not malicious on volume alone; none were tied to a post-compromise command-and-control pattern.

Export | Show empty columns | 2718 items | Search | 00:02.8

Filters: Add filter

| TimeGenerated | InitiatingProcessFil... | InitiatingProcessCo... | RemoteIP | RemotePort | DeviceName |
|---|---|---|---|---|---|
| Oct 9, 2026 2:41:1... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:40:5... | svchost.exe | svchost.exe -k NetworkS... | 80.66.83.43 | 65118 | valued-customer |
| Oct 9, 2026 2:37:5... | ntoskrnl.exe | | ::ffff:1.14.149.228 | 63702 | valued-customer |
| Oct 9, 2026 2:36:5... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:36:1... | ntoskrnl.exe | | ::ffff:198.235.24.222 | 62146 | valued-customer |
| Oct 9, 2026 2:35:0... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:33:0... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:32:3... | ntoskrnl.exe | | 85.217.149.13 | 52052 | valued-customer |
| Oct 9, 2026 2:32:0... | onedrive.exe | "OneDrive.exe" /backgr... | 4.150.223.110 | 443 | valued-customer |
| Oct 9, 2026 2:28:4... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:28:1... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:28:0... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:26:4... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:22:0... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:19:3... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:16:2... | | | 20.89.1.9 | 443 | valued-customer |
| Oct 9, 2026 2:16:2... | | | 13.69.116.104 | 443 | valued-customer |
| Oct 9, 2026 2:16:2... | | | 20.89.1.9 | 443 | valued-customer |
| Oct 9, 2026 2:16:2... | | | 13.69.116.104 | 443 | valued-customer |
| Oct 9, 2026 2:16:2... | svchost.exe | svchost.exe -k utcsvc -p | 20.89.1.9 | 443 | valued-customer |
| Oct 9, 2026 2:16:2... | svchost.exe | svchost.exe -k utcsvc -p | 13.69.116.104 | 443 | valued-customer |
| Oct 9, 2026 2:16:0... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:10:1... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:09:5... | | | 168.63.129.16 | 53 | valued-customer |
| Oct 9, 2026 2:09:5... | mysqld.exe | "mysqld.exe" "--defaults... | ::ffff:35.203.211.65 | 56443 | valued-customer |
| Oct 9, 2026 2:02:0... | onedrive.exe | "OneDrive.exe" /backgr... | 20.184.175.13 | 443 | valued-customer |
| Oct 9, 2026 2:01:0... | | | 20.25.227.174 | 443 | valued-customer |
| Oct 9, 2026 2:01:0... | nissrv.exe | NisSrv.exe | 20.25.227.174 | 443 | valued-customer |
| Oct 9, 2026 2:00:3... | | | 10.3.0.33 | | valued-customer |
| Oct 9, 2026 2:00:1... | | | 10.3.0.33 | | valued-customer |

*Figure 15. DeviceNetworkEvents since exposure (2,718 items).*

Add filter

| SrcIp | DestIp | count_ |
|---|---|---|
| | 10.2.0.38 | 54155 |
| 10.0.0.8 | 10.0.0.117 | 46188 |
| 10.0.0.8 | 10.2.0.32 | 28111 |
| | 10.0.0.155 | 17309 |
| | 10.0.0.198 | 17101 |
| | 10.0.0.8 | 13729 |
| | 10.2.0.30 | 13603 |
| 10.0.0.8 | 10.3.0.18 | 12314 |
| | 10.3.0.43 | 11608 |
| | 10.3.0.24 | 10797 |
| | 10.0.0.103 | 10515 |
| | 10.3.0.42 | 10132 |
| | 10.2.0.26 | 10021 |
| 10.0.0.8 | 10.0.0.119 | 9961 |
| | 10.2.0.36 | 9959 |
| | 10.3.0.54 | 9786 |
| | 10.3.0.23 | 9483 |
| | 10.3.0.33 | 9440 |
| | 10.0.0.10 | 9047 |
| | 10.0.0.115 | 8912 |

**Network flow analytics.** Flows are concentrated inside private 10.0.0.0/8 space: 10.0.0.8 to 10.0.0.117 (46,188) and to 10.2.0.32 (28,111) lead the list.

Rows with a blank source are flows where the source is not parsed in this summary view (for example 10.2.0.38 with 54,155).

**Result:** heavy but stable internal VNet traffic; no unexplained external destination appears among the top flows.

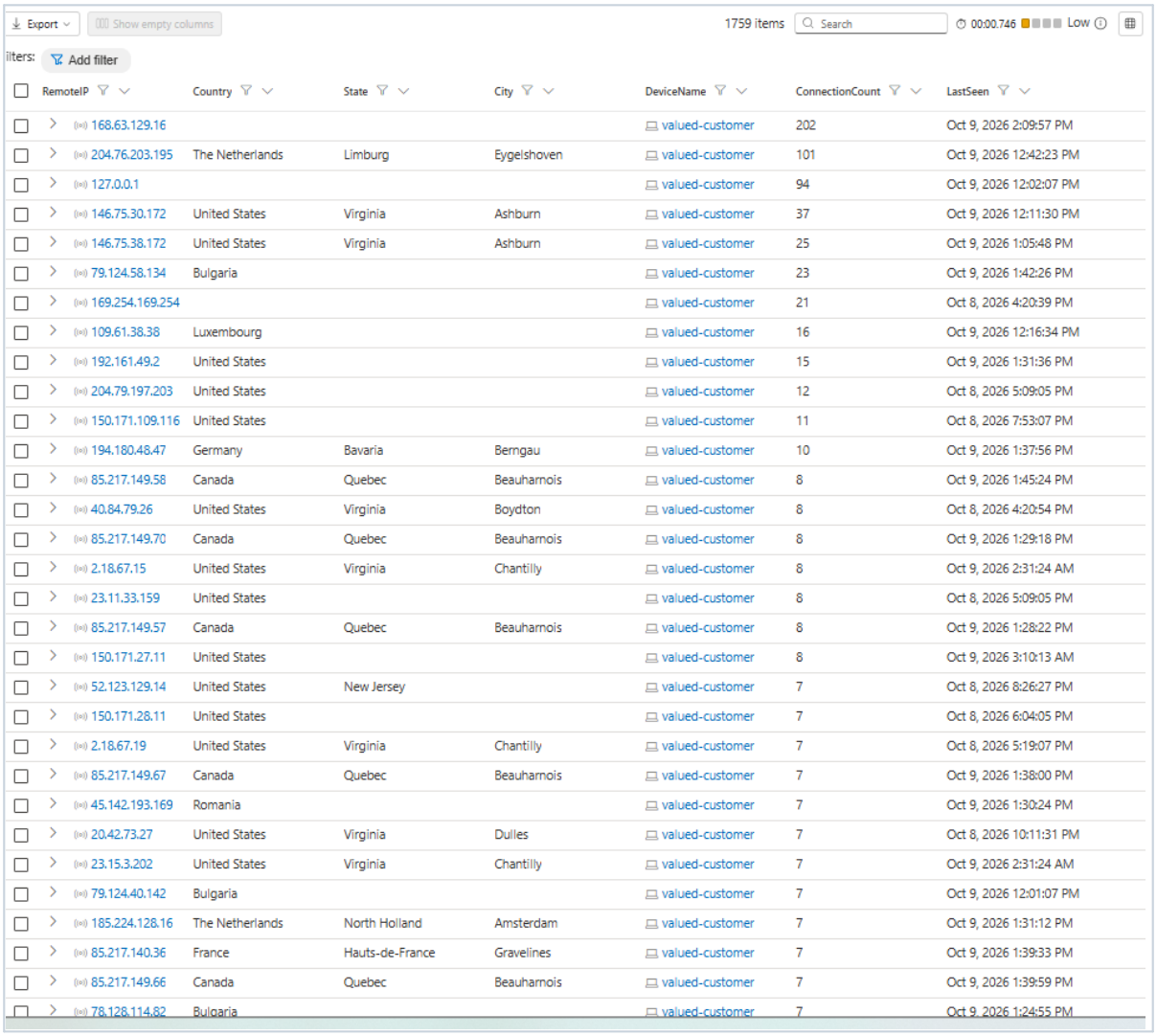
*Figure 16. NTANetAnalytics - top 20 flows by connection count.*

## 4.4 Geographic enrichment of external traffic

Private ranges were stripped from DeviceNetworkEvents and each remaining address was enriched with `geo_info_from_ip_address()`, then ranked by connection count.

**Geo-enrichment query**

```
DeviceNetworkEvents
| where DeviceName startswith MyDevice and isnotempty(RemoteIP)
| where not(ipv4_is_private(RemoteIP))
| extend GeoInfo = geo_info_from_ip_address(RemoteIP)
| extend Country = tostring(parse_json(GeoInfo).country), State = tostring(parse_json(GeoInfo).state),
         City = tostring(parse_json(GeoInfo).city)
| summarize ConnectionCount = count(), LastSeen = max(TimeGenerated) by RemoteIP, Country, State, City, DeviceName
| order by ConnectionCount desc
```



Export | Show empty columns | 1759 items | Search | 00:00.746 | Low

Filters: Add filter

| RemoteIP | Country | State | City | DeviceName | ConnectionCount | LastSeen |
|---|---|---|---|---|---|---|
| 168.63.129.16 | | | | valued-customer | 202 | Oct 9, 2026 2:09:57 PM |
| 204.76.203.195 | The Netherlands | Limburg | Eygelshoven | valued-customer | 101 | Oct 9, 2026 12:42:23 PM |
| 127.0.0.1 | | | | valued-customer | 94 | Oct 9, 2026 12:02:07 PM |
| 146.75.30.172 | United States | Virginia | Ashburn | valued-customer | 37 | Oct 9, 2026 12:11:30 PM |
| 146.75.38.172 | United States | Virginia | Ashburn | valued-customer | 25 | Oct 9, 2026 1:05:48 PM |
| 79.124.58.134 | Bulgaria | | | valued-customer | 23 | Oct 9, 2026 1:42:26 PM |
| 169.254.169.254 | | | | valued-customer | 21 | Oct 8, 2026 4:20:39 PM |
| 109.61.38.38 | Luxembourg | | | valued-customer | 16 | Oct 9, 2026 12:16:34 PM |
| 192.161.49.2 | United States | | | valued-customer | 15 | Oct 9, 2026 1:31:36 PM |
| 204.79.197.203 | United States | | | valued-customer | 12 | Oct 8, 2026 5:09:05 PM |
| 150.171.109.116 | United States | | | valued-customer | 11 | Oct 8, 2026 7:53:07 PM |
| 194.180.48.47 | Germany | Bavaria | Berngau | valued-customer | 10 | Oct 9, 2026 1:37:56 PM |
| 85.217.149.58 | Canada | Quebec | Beauharnois | valued-customer | 8 | Oct 9, 2026 1:45:24 PM |
| 40.84.79.26 | United States | Virginia | Boydton | valued-customer | 8 | Oct 8, 2026 4:20:54 PM |
| 85.217.149.70 | Canada | Quebec | Beauharnois | valued-customer | 8 | Oct 9, 2026 1:29:18 PM |
| 2.18.67.15 | United States | Virginia | Chantilly | valued-customer | 8 | Oct 9, 2026 2:31:24 AM |
| 23.11.33.159 | United States | | | valued-customer | 8 | Oct 8, 2026 5:09:05 PM |
| 85.217.149.57 | Canada | Quebec | Beauharnois | valued-customer | 8 | Oct 9, 2026 1:28:22 PM |
| 150.171.27.11 | United States | | | valued-customer | 8 | Oct 9, 2026 3:10:13 AM |
| 52.123.129.14 | United States | New Jersey | | valued-customer | 7 | Oct 8, 2026 8:26:27 PM |
| 150.171.28.11 | United States | | | valued-customer | 7 | Oct 8, 2026 6:04:05 PM |
| 2.18.67.19 | United States | Virginia | Chantilly | valued-customer | 7 | Oct 8, 2026 5:19:07 PM |
| 85.217.149.67 | Canada | Quebec | Beauharnois | valued-customer | 7 | Oct 9, 2026 1:38:00 PM |
| 45.142.193.169 | Romania | | | valued-customer | 7 | Oct 9, 2026 1:30:24 PM |
| 20.42.73.27 | United States | Virginia | Dulles | valued-customer | 7 | Oct 8, 2026 10:11:31 PM |
| 23.15.3.202 | United States | Virginia | Chantilly | valued-customer | 7 | Oct 9, 2026 2:31:24 AM |
| 79.124.40.142 | Bulgaria | | | valued-customer | 7 | Oct 9, 2026 12:01:07 PM |
| 185.224.128.16 | The Netherlands | North Holland | Amsterdam | valued-customer | 7 | Oct 9, 2026 1:31:12 PM |
| 85.217.140.36 | France | Hauts-de-France | Gravelines | valued-customer | 7 | Oct 9, 2026 1:39:33 PM |
| 85.217.149.66 | Canada | Quebec | Beauharnois | valued-customer | 7 | Oct 9, 2026 1:39:59 PM |
| 78.128.114.82 | Bulgaria | | | valued-customer | 7 | Oct 9, 2026 1:24:55 PM |

*Figure 17. 1,759 unique external IPs, ranked by connection count with country, state and city.*

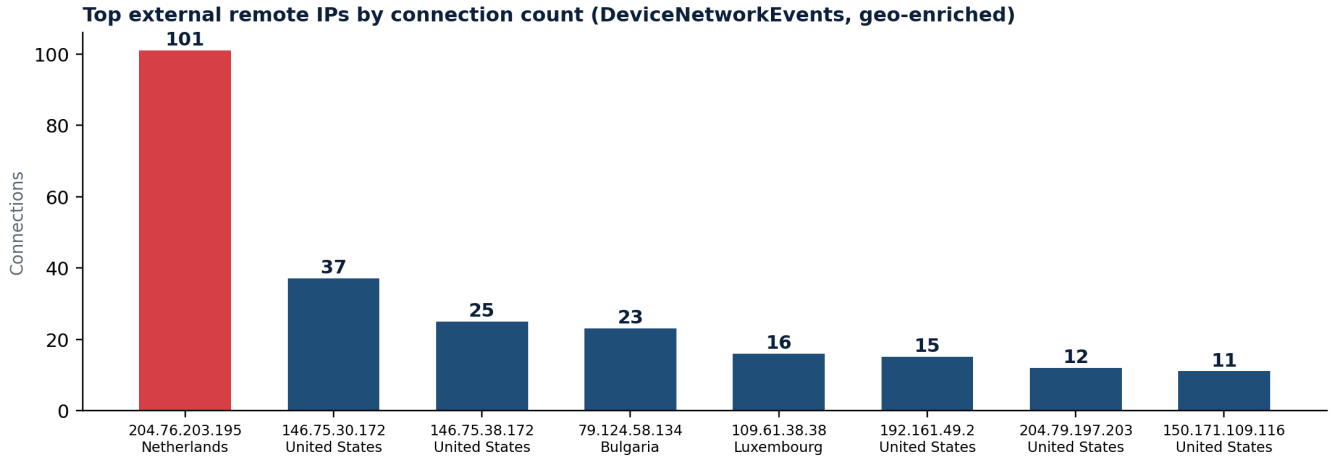



*Figure 18. Top external IPs by connection count (infrastructure and loopback addresses such as 168.63.129.16, 127.0.0.1 and 169.254.169.254 excluded).*

**Reading the result.** Geography confirms global reach - the Netherlands, United States, Bulgaria, Luxembourg, Germany, Canada, Romania and France all appear. The top address, 204.76.203.195 (Netherlands, 101 connections),

is the busiest non-infrastructure peer and is listed for follow-up. Some listed addresses belong to cloud and CDN providers; geolocation shows where an IP is registered, not who is behind it.

## 4.5 Microsoft Sentinel visualisation

The geo-enrichment was repeated on the MySQL audit log itself, extracting the source IP from RawData, dropping loopback entries and keeping only IPs with resolved coordinates. A Sentinel workbook then plotted inbound authentication origins: bubble size equals attempts, colour runs from red (failures) to green (success).

**Sentinel map query (condensed)**

```
MySQLAudit_CL
| where TimeGenerated > ServerVulnerableDateTime
| extend DeviceName = tostring(split(_ResourceId, "/")[-1])
| extend ExtractedIp = extract(@"@'?([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+)'?", 1, RawData)
| where isnotempty(ExtractedIp) and ExtractedIp != "127.0.0.1" and ExtractedIp != "::1"
| extend GeoInfo = geo_info_from_ip_address(ExtractedIp)
| extend Country = tostring(parse_json(GeoInfo).country), Latitude = tostring(parse_json(GeoInfo).latitude),
        Longitude = tostring(parse_json(GeoInfo).longitude)
| where isnotempty(Latitude) and isnotempty(Longitude)
| summarize AttemptCount = count() by ExtractedIp, Country, State, City, Latitude, Longitude, DeviceName
| order by AttemptCount desc
```

Results | Query history | Getting started

Export | Show empty columns | 63 items | Search | 00:03.268 | Low

Filters: Add filter

| ExtractedIp | Country | State | City | Latitude | Longitude | AttemptCount | DeviceName |
|---|---|---|---|---|---|---|---|
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 180 | corp-vi26-t340 |
| 77.90.185.21 | Germany | | | 51.2993 | 9.491 | 178 | corp-vi26-t340 |
| 64.89.163.94 | United States | | | 37.751 | -97.822 | 71 | corp-nao1-lreed |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 70 | corpna1da |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 70 | corp-ad127-na274 |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 63 | corp-sat-vinder |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 63 | corp-nao1-lreed |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 42 | gpu-farm-eastus |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 42 | corp-na04-fa566a |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 42 | valued-customers100 |
| 64.89.163.178 | United States | | | 37.751 | -97.822 | 38 | corpna1da |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 35 | corp-oscorp |
| 77.90.185.30 | Germany | | | 51.2993 | 9.491 | 35 | corp-na01-feda1 |
| 64.89.163.94 | United States | | | 37.751 | -97.822 | 16 | corp-ad127-na274 |
| 77.90.185.21 | Germany | | | 51.2993 | 9.491 | 14 | corp-ra01-te123da |
| 77.90.185.21 | Germany | | | 51.2993 | 9.491 | 14 | corp-nao1-lreed |
| 77.90.185.21 | Germany | | | 51.2993 | 9.491 | 14 | corp-sat-vinder |
| 77.90.185.21 | Germany | | | 51.2993 | 9.491 | 14 | corp-ad127-na274 |
| 45.128.199.171 | The Netherlands | North Holland | Amsterdam | 52.3716 | 4.8883 | 12 | corp-nao1-lreed |
| 64.89.163.94 | United States | | | 37.751 | -97.822 | 11 | corp-sat-vinder |

*Figure 19. Map query results. The workspace is shared, so rows for other device names are out of scope; the row for valued-customers1000 shows 42 attempts from 77.90.185.30 (Germany).*

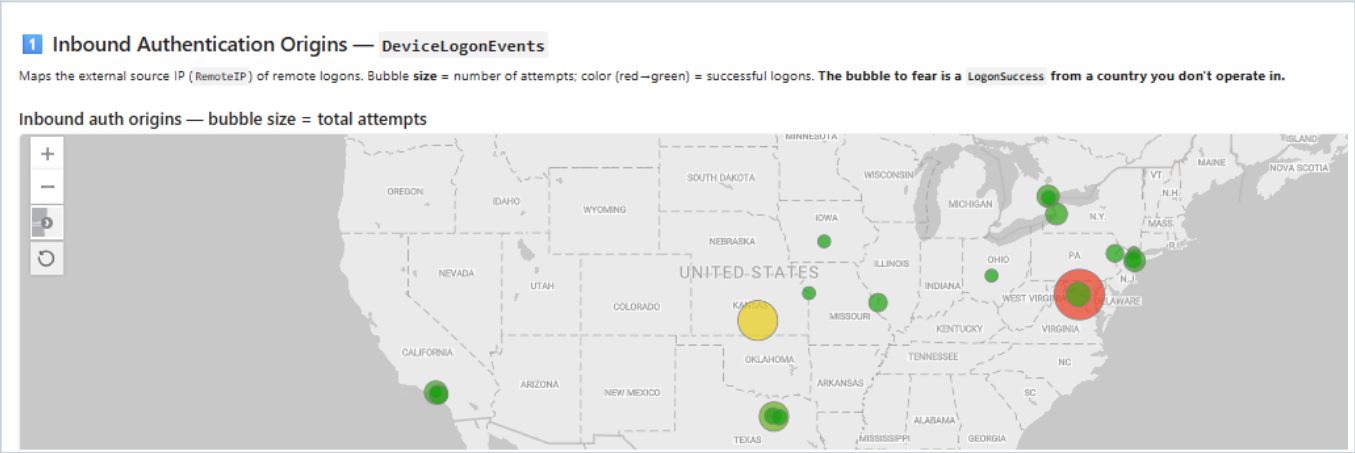



*Figure 20. Sentinel workbook: inbound authentication origins, sized by attempts.*

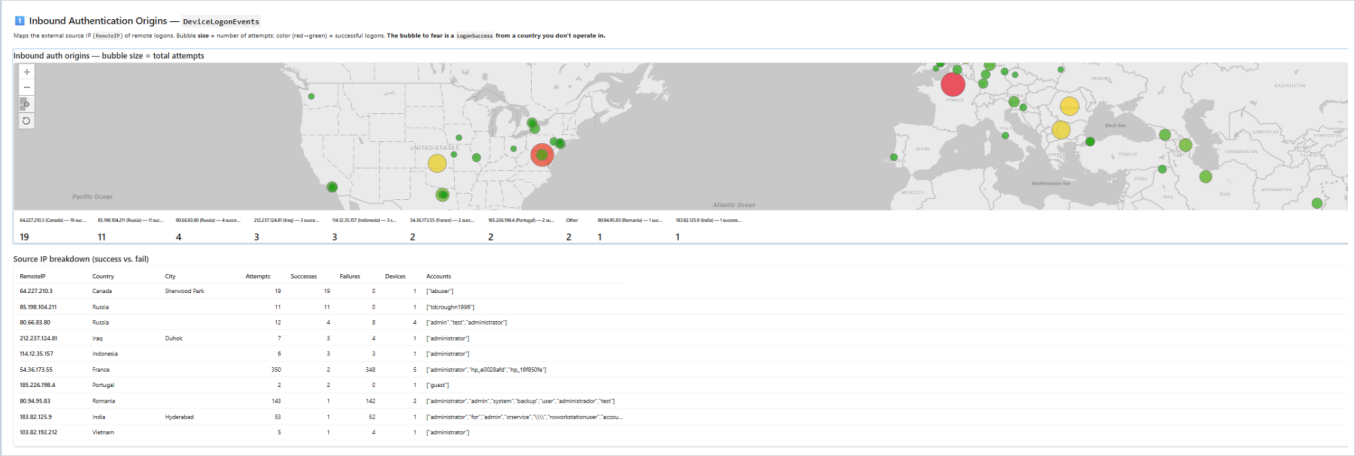



*Figure 21. Workbook (wide view) with the source-IP breakdown of successes versus failures.*

## 4.6 Consolidated analysis

Putting the sources together gives a single narrative (the attack chain is shown in Section 1). The timeline below shows how the events sit in sequence.

*Times in UTC; spacing is sequential, not to scale (Oct 9 timestamps from MySQL audit RawData and MDE telemetry)*

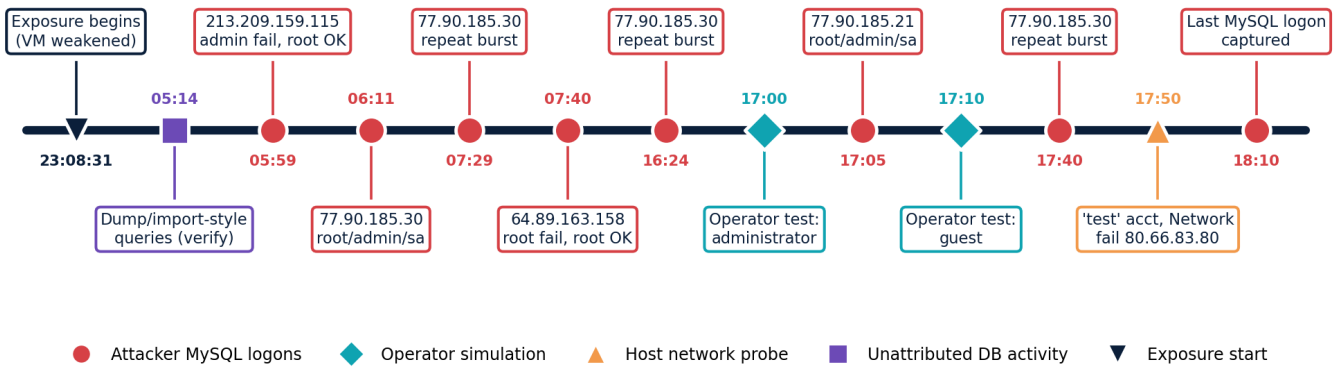



*Figure 22. Incident timeline (sequential view).*

## 4.7 Indicators of compromise

| Indicator | Type | Context |
|---|---|---|
| 77.90.185.30 | IPv4 (geo: Germany) | Most active source; root/admin/sa bursts repeated through the window; 42 attempts against valued-customers1000 in the workbook |
| 77.90.185.21 | IPv4 (geo: Germany) | root/admin/sa burst at ~17:05Z followed by root success |
| 213.209.159.115 | IPv4 | First captured intrusion: admin failure then root success at ~05:59Z |
| 64.89.163.158 | IPv4 (geo: United States) | root failure then root success, ~07:40Z; session issued SET AUTOCOMMIT=0 |
| 45.128.199.171 | IPv4 (geo: Netherlands) | Appears in the exported MySQL audit dataset |
| 80.66.83.80 | IPv4 | Failed Network logon for account 'test' on the host (~17:50Z) |
| 204.76.203.195 | IPv4 (geo: Netherlands) | Highest-volume non-infrastructure peer in DeviceNetworkEvents (101); follow-up only, not tied to MySQL |
| root, admin, sa, test | Usernames | Credential-guessing dictionary observed |

### MITRE ATT&CK mapping

| Technique | ID | Evidence |
|---|---|---|
| Active Scanning / Exploit Public-Facing Application (exposure) | T1595 / T1190 | Internet sources located and engaged an open MySQL service |
| Brute Force: password guessing / spraying | T1110 | Rapid root, admin, sa attempts producing Access denied |
| Valid Accounts: Default Accounts | T1078.001 | Successful login as the built-in root account |

# 5. Containment (Phase 8)

**PHASE 8** NIST SP 800-61: Containment, Eradication & Recovery

Containment limited further exposure while preserving evidence for the record.

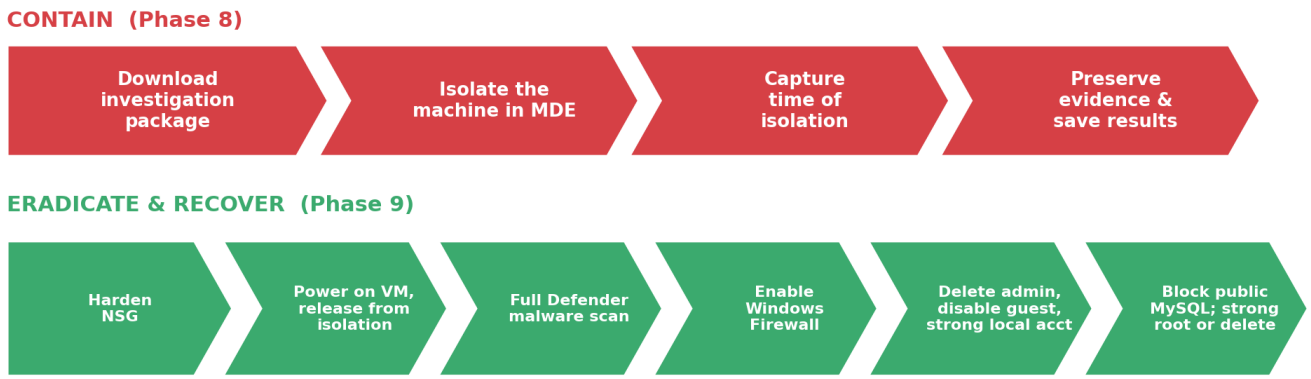



*Figure 23. Response workflow for Phases 8 and 9.*

| Step | Action | Why |
|---|---|---|
| 1 | Downloaded the MDE **investigation package** from the device page | Preserves a forensic snapshot (processes, services, network, firewall rules, local accounts) before the machine changes |
| 2 | **Isolated the machine** in Microsoft Defender for Endpoint | Cuts network access while retaining the MDE management channel, stopping further attacker interaction |
| 3 | **Captured the time of isolation** | Needed to measure time to contain and to bound the incident window. *Value not recorded in the source notes - insert here:* _____ |

Time metrics: exposure at 23:08:31Z on 8 Oct; first captured attacker success at ~05:59Z on 9 Oct (about 6 h 50 min later, based on the captured result set); last captured attacker logon ~18:10Z on 9 Oct. Time to contain depends on the isolation timestamp noted above.

# 6. Eradication and Recovery (Phase 9)

**PHASE 9** NIST SP 800-61: Containment, Eradication & Recovery

Eradication removed the weaknesses that made the intrusion possible; recovery returned the VM to service in a hardened state.

| Control | Before (exposed) | After (hardened) |
|---|---|---|
| Azure NSG | Wide open (ingress and egress) | Hardened - restricted to required sources only |
| Windows Firewall | Disabled | Enabled (wf.msc) |
| administrator account | Enabled, weak | Deleted |
| guest account | Enabled, weakened password | Disabled |
| Local logon account | Weak / default | Strong username and password |
| MySQL public access | Reachable from the internet | Logon from the public internet not allowed |
| MySQL root (network) | Weak password | Strong password, or account deleted |
| Malware status | Unknown | Full Windows Defender scan run |
| Machine state | Isolated | Powered on and released from isolation |

### Recovery verification checklist

Confirm the full Defender scan completed clean and no new alerts are open on the device.

Re-run Valued_MYSQLSERVER_Login and Valued_VM_Login for 24-48 hours after release and confirm no new external success events.

Confirm port 3306 is unreachable from the internet (external test) and that MySQL `root` cannot log in remotely.

# 7. Post-Incident Activity (Phase 10)

**PHASE 10** NIST SP 800-61: Post-Incident Activity

## 7.1 What went well

- The log pipeline (MDE, DCR to MySQLAudit_CL, Sentinel) produced evidence for every stage of the incident.
- Connection-ID pairing turned raw general-log text into clean LogonSuccess/LogonFailure events.
- Simulated authentication proved the host detection before real traffic arrived.
- Evidence was preserved (investigation package) before isolation and recovery.

## 7.2 Lessons learned

| Lesson | Detail |
|---|---|
| Exposure is found fast | An unprotected database port was located and attacked within hours; scanners do not need a reason to target you. |
| Default accounts are the first target | root, admin and sa were tried in sequence; sensible defaults and removal of default accounts defeat the dictionary. |
| Name mismatches break queries | MDE truncates device names to 15 characters; == is exact and case-sensitive, so use the exact name or `startswith`. Different tables use different name forms. |
| Zero results is not zero risk | A rule returns nothing until the behaviour occurs. Test rules by simulating the activity. |
| Know your time zones | Portal screenshots show local time (EDT) while MySQL RawData is UTC. Misreading AM/PM or offsets can reorder an incident timeline. |
| Absence of host artifacts is not absence of impact | Root held full database privilege; MySQL-level persistence does not necessarily appear in host telemetry. |

## 7.3 Recommendations

| Priority | Recommendation |
|---|---|
| **CRITICAL** | Never expose TCP 3306 to the internet. Bind MySQL to private interfaces, use Private Endpoint/VPN/Bastion, and keep NSG inbound rules deny-by-default. |
| **CRITICAL** | Remove default/built-in accounts, forbid remote *root*, enforce strong unique credentials and least-privilege application users. |
| **HIGH** | Promote the two rules to Sentinel analytics: alert when one IP tries several distinct usernames in under a minute, and when a first-seen external IP succeeds. Add entity mapping for IP and account. |
| **HIGH** | Review MySQL internals after any root compromise: mysql.user and grants, scheduled events, triggers, UDFs, and any `INTO OUTFILE` or `LOAD DATA` in the general log. |
| **MEDIUM** | Require TLS for connections, forward Windows Security events, and extend log retention to cover investigations. |
| **MEDIUM** | Record detection, containment and recovery timestamps in a template so MTTD and MTTR can be reported. |
| **LOW** | Put the Defender investigation-package and isolation steps into a playbook so containment is repeatable. |

## 7.4 Analyst notes and limitations

- **Unattributed block at 05:14Z.** The *world* schema statements resemble a Workbench dump import (SET FOREIGN_KEY_CHECKS, DROP TABLE IF EXISTS, CREATE TABLE, INSERT, UNLOCK TABLES) and predate the first captured attacker login. Correlate the connection ID and source IP to confirm whether this was operator activity.
- **Scope of "no persistence".** The conclusion covers the MDE tables reviewed, not MySQL-internal changes.
- **Counts differ slightly** between portal screenshots and CSV exports (for example registry 1,162 vs 1,114) because of ingestion timing; portal values are used in charts.
- **The network-events narrative** in the original notes duplicated the registry-events text; the analysis here was rewritten from the screenshot.
- **Geolocation** reflects IP registration and is not attribution. Some listed IPs belong to cloud or CDN providers.
- **Placeholders:** incident reference, analyst name and time of isolation were not in the source and must be completed.
- **LLM-assisted triage** was used on exported results; the findings above were checked against the screenshots but should be validated before external distribution.

# Appendix A - Hunting query catalogue

All post-exposure queries share the same header and differ in table and projected columns.

**Common template**

```
let FirstSuccessfulLogonTime = todatetime("2026-10-08T23:08:31.8034371Z");
let MyDevice = "valued-customer";
<TABLE>
| where TimeGenerated >= FirstSuccessfulLogonTime
| where DeviceName == MyDevice
| project <COLUMNS>
| order by TimeGenerated desc
```

| Table | Columns projected |
|---|---|
| DeviceLogonEvents | TimeGenerated, AccountName, DeviceName, ActionType, LogonType, RemoteIP |
| DeviceProcessEvents | TimeGenerated, AccountName, ProcessCommandLine, InitiatingProcessCommandLine, FolderPath, FileName, DeviceName |
| DeviceFileEvents | TimeGenerated, InitiatingProcessAccountName, FolderPath, FileName, ActionType, DeviceName |
| DeviceRegistryEvents | TimeGenerated, InitiatingProcessCommandLine, RegistryKey, RegistryValueName, DeviceName |
| DeviceNetworkEvents | TimeGenerated, InitiatingProcessFileName, InitiatingProcessCommandLine, RemoteIP, RemotePort, DeviceName |

**NTANetAnalytics top flows**

```
NTANetAnalytics
| where TimeGenerated >= VulnerableDateTime
| summarize count() by SrcIp, DestIp
| order by count_ desc
| take 20
```

**Discovery and troubleshooting queries**

```
DeviceInfo
| where DeviceName startswith "valued"
| project DeviceId, DeviceName, OSPlatform, OSVersion

MySQLAudit_CL | getschema
MySQLAudit_CL | project TimeGenerated, RawData, _ResourceId
MySQLAudit_CL | where _ResourceId contains "Valued-customers"
              | distinct _ResourceId | project DeviceName = split(_ResourceId, "/")[-1]

DeviceLogonEvents
| where DeviceName == "valued-customer"
| summarize count() by AccountName, ActionType, LogonType
| order by count_ desc
```

# Appendix B - Evidence index

| Evidence | Source | Section |
|---|---|---|
| Detection validation screenshots (administrator, guest) | Defender Advanced Hunting | 3.5 |
| MySQL logon results (Query 1) | Log Analytics, MySQLAudit_CL | 4.2 |
| Username targeting and logon action types | Charts built from audit export | 4.2 |

| Evidence | Source | Section |
|---|---|---|
| MySQL query tracking (Query 2) | Log Analytics, MySQLAudit_CL | 4.3.1 |
| Honeypot logon events (Query 3) | DeviceLogonEvents | 4.3.2 |
| Logon, process, file, registry, network screenshots | Defender Advanced Hunting | 4.3.3 |
| NTANetAnalytics top flows | Log Analytics | 4.3.3 |
| Geo-enriched connections and top IPs | DeviceNetworkEvents | 4.4 |
| Sentinel query results and workbook maps | Log Analytics / Sentinel | 4.5 |
| Attack chain, timeline, lifecycle, architecture, workflow | Diagrams drawn for this report | 1, 2, 4.6, 5 |